Robustness of EEG Feature Representations Under Cross-Dataset Distribution Shift Using Classical Models

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Abstract

EEG-based brain-computer interfaces (BCIs) frequently report high within-dataset classification accuracy, yet performance collapses when models are applied to new datasets recorded under different conditions. This paper investigates whether frequency-domain feature representations are more robust than time-domain statistical features under cross-dataset distribution shift. Using two publicly available motor imagery datasets—PhysioNet EEG Motor Imagery (109 subjects) and BCI Competition IV Dataset 2a (9 subjects)—we evaluate 9 feature-model combinations under strict zero-shot transfer in both directions. FFT magnitude features showed the smallest mean normalised generalisation gap (0.065), followed by band power (0.070) and time-domain features (0.071), though differences were modest. Critically, within-dataset accuracy was near chance for PhysioNet (53–56%), indicating that commonly reported high within-dataset accuracies do not necessarily reflect genuine cross-dataset generalisation. A $9\times$ amplitude difference between recording systems was empirically confirmed as the primary barrier to cross-dataset transfer; log-transformed frequency features partially mitigate this shift.

I. Introduction

Brain-computer interfaces (BCIs) based on electroencephalography (EEG) have shown considerable promise for enabling motor-impaired individuals to communicate through imagined movement. Motor imagery—the mental rehearsal of a motor action—produces characteristic suppression of the mu (8–12 Hz) and beta (13–30 Hz) rhythms over the contralateral motor cortex, providing a reliable neural signal for binary classification.

Despite decades of research, EEG-based BCIs face a persistent barrier: models trained on one dataset rarely transfer to another. Studies routinely report within-dataset accuracies of 80–90% [13, 8], yet these numbers are produced by training and testing on the same subjects under the same recording conditions. When the same model is applied to a different dataset—different subjects, amplifiers, electrode configurations—performance collapses to near chance. This gap between reported and real-world accuracy is a fundamental challenge for clinical BCI deployment.

The causes are multiple: electrode impedance and amplifier gain differences introduce amplitude-scale shifts; different subject pools introduce inter-subject variability; different experimental protocols introduce trial-timing differences. What remains underexplored is *which feature representation* degrades least under these conditions, and why.

This paper addresses that question directly. We compare three classical feature representations—time-domain statistics, band power, and FFT magnitude—under strict zero-shot cross-dataset transfer. Using classical machine learning deliberately isolates the feature representation as the variable of interest. Deep learning conflates feature extraction with classification, making it impossible to attribute generalisation differences to the representation itself.

Our central hypothesis is that frequency-domain features will be more robust under distribution shift than time-domain features, because EEG's oscillatory band structure is more stable across recording conditions than its absolute amplitude.

Contributions:

- A controlled empirical comparison of three classical feature representations under strict cross-dataset transfer in both directions.
- Empirical quantification of the amplitude-scale distribution shift between two widely used public datasets.
- A transparent within-dataset accuracy characterisation for an unscreened subject population, providing a realistic pooled trial-wise baseline for interpreting cross-dataset robustness.

II. Related Work

A. Motor Imagery Classification

The classification of motor imagery EEG has been studied since Pfurtscheller and Neuper [10] established the neurophysiological basis of event-related desynchronisation in the mu and beta bands. Standard approaches extract band-power features using Welch's PSD estimation and classify with linear discriminant analysis or SVMs. Common spatial patterns (CSP) and its variants have been widely applied. Within-dataset accuracies of 70–90% are regularly reported, particularly on competition benchmarks such as BCI Competition IV Dataset 2a.

B. Cross-Dataset Generalisation

Cross-dataset generalisation is significantly less studied than within-dataset evaluation. Jayaram et al. [5] demonstrated

through the MOABB benchmark that many published BCI algorithms show dramatic performance drops under cross-session and cross-subject evaluation [14, 4], explicitly framing this as a *cross-dataset variability problem* that weakens generalisation.

Crucially, the part of the literature that most directly addresses "robustness across datasets" has mostly targeted deep-learning and domain adaptation approaches: recent reviews of MI-EEG classification are dominated by deep architectures [12], while contemporary cross-dataset MI work centres on transfer learning, pre-training, covariate-shift alignment, and graph-based models [7]. Classical feature-engineering work exists but is almost exclusively evaluated *within* a single dataset [9].

C. Gap This Study Fills

To our knowledge, prior work has rarely isolated feature representation as the *sole* variable under a strict train-on-A, test-on-B design with identical preprocessing and both transfer directions. Existing work more often focuses on domain adaptation, deep architectures, or cross-subject evaluation within a single dataset. Our study provides a controlled classical baseline against which such approaches can be compared.

III. Methods

A. Datasets

PhysioNet EEG Motor Imagery [11, 3] comprises recordings from 109 subjects at 160 Hz on 64 channels. Only the left/right motor imagery runs (runs 0, 1, and 2 in MOABB notation) were retained. After preprocessing, 4,918 trials were kept (2,480 left, 2,438 right).

BCI Competition IV Dataset 2a [2], accessed as `BNCI2014_001` via MOABB [6], comprises recordings from 9 subjects at 250 Hz on 22 EEG channels. Only left-hand (event 769) and right-hand (event 770) trials were retained, yielding 2,592 trials (1,296 each).

B. Preprocessing

An identical pipeline was applied to both datasets:

1. **Channel selection**: both recordings restricted to the 22 BCI2a channels (a strict subset of PhysioNet's 64). No interpolation required.
2. **Resampling**: PhysioNet upsampled from 160 Hz to 250 Hz using MNE polyphase resampling. Upsampling was preferred to preserve all frequency content below the original Nyquist (80 Hz).
3. **Bandpass filtering**: 4–40 Hz FIR filter; captures the motor-relevant mu and beta rhythms while removing DC drift and EMG artefacts.
4. **Epoch extraction**: 0.5–2.5 s post-cue; the 0.5 s offset avoids the cue-evoked visual potential. No baseline correction applied (bandpass renders epoch means approximately zero).
5. **Label encoding**: left $= 0$, right $= 1$.

Output arrays have shape $(\text{n_trials}, 22, 501)$.

C. Feature Extraction

Feature A — Time Domain (44-dim). Mean and variance per channel over time. Because the signal is bandpass-filtered, means are approximately zero; variance is the primary discriminative signal. This set is intentionally simple—it tests the weakest possible time-domain representation, not time-domain features in general.

Feature B — Band Power (88-dim). Welch PSD (nperseg $= 256$, 50% overlap, Hann window) per channel; average power in delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), and beta (13–30 Hz). Natural log transform applied to normalise the skewed distribution and compress amplitude differences across recording systems. Yields $22 \times 4 = 88$ features.

Feature C — FFT Magnitude (792-dim). FFT magnitude restricted to 4–40 Hz, linearly interpolated to 36 fixed frequency bins per channel. No log transform; StandardScaler inside the pipeline handles scale. Linear interpolation (over cubic spline) was chosen to preserve sharp spectral peaks. Yields $22 \times 36 = 792$ features.

D. Classifiers

Three classifiers were evaluated in scikit-learn Pipelines:

- **Logistic Regression**: L2 regularisation, max_iter $= 5000$, preceded by StandardScaler.
- **SVM**: RBF kernel (C=1, gamma = `scale`), preceded by StandardScaler.
- **Random Forest**: 100 trees; no feature scaling (scale-invariant).

Placing StandardScaler inside the Pipeline ensures it is fitted only on training data, preventing leakage.

E. Evaluation Protocol

Within-dataset evaluation used 5-fold stratified cross-validation (shuffle, random_state $= 42$). Because trials were pooled across subjects before splitting, these scores should be interpreted as pooled trial-wise baselines rather than fully subject-independent grouped-subject estimates.

Cross-dataset evaluation used zero-shot transfer: train on the full source dataset, test on the full target dataset. Both directions were evaluated: PhysioNet $\rightarrow$ BCI2a and BCI2a $\rightarrow$ PhysioNet.

Primary metric: *balanced accuracy* (mean per-class recall), robust to class imbalance. All experiments used seed 42.

The normalised generalisation gap is defined as:

$$\text{Norm. Gap} = \frac{BA_{\text{within}} - BA_{\text{cross}}}{BA_{\text{within}}} \tag{1}$$

An interpretation guide adapted from cross-dataset MI literature: gap < 0.10 is small degradation; 0.10–0.20 is moderate; > 0.20 is severe.

Table 1. Within-dataset balanced accuracy (mean ± std, 5-fold CV). Chance level is 0.500.

Feature	Model	PhysioNet	BCI2a
Time Domain	Log. Reg.	0.535 ± 0.018	0.639 ± 0.014
	SVM	0.523 ± 0.010	0.591 ± 0.029
	Rand. For.	0.509 ± 0.007	0.588 ± 0.038
Band Power	Log. Reg.	0.559 ± 0.026	0.642 ± 0.027
	SVM	0.529 ± 0.015	0.596 ± 0.021
	Rand. For.	0.512 ± 0.017	0.608 ± 0.019
FFT	Log. Reg.	0.556 ± 0.014	0.613 ± 0.015
	SVM	0.539 ± 0.010	0.595 ± 0.012
	Rand. For.	0.520 ± 0.024	0.567 ± 0.010

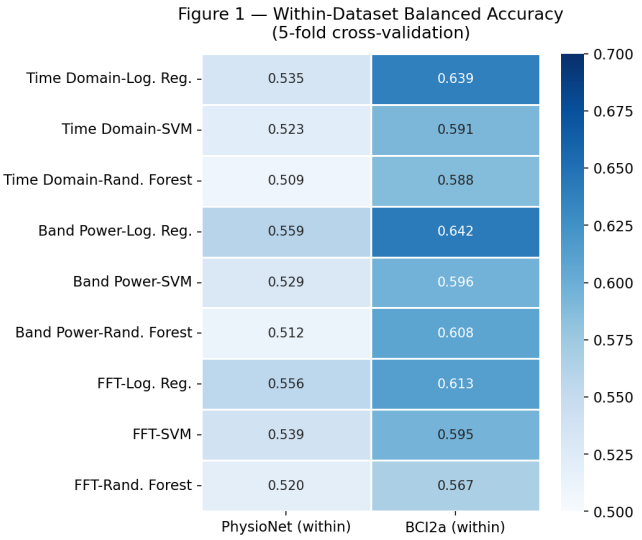

Figure 1. Within-dataset balanced accuracy heatmap (5-fold CV). BCI2a consistently outperforms PhysioNet; all PhysioNet values are near chance.

IV. Results

A. Within-Dataset Accuracy

Table 1 reports balanced accuracy for within-dataset evaluation. BCI2a accuracy (59–64%) consistently exceeded PhysioNet (51–56%) across all feature-model combinations. Band Power with Logistic Regression achieved the highest single result: 64.2% on BCI2a. Figure 1 visualises this as a heatmap.

PhysioNet accuracy was near chance across all configurations, reflecting the unscreened 109-subject pool and high inter-subject variability in motor imagery signal quality.

B. Cross-Dataset Accuracy and Generalisation Gap

Table 2 and Figure 3 report cross-dataset results. The two transfer directions showed strikingly different patterns.

PhysioNet → BCI2a: gaps were near zero or negative across all feature sets. Logistic Regression achieved slightly *higher* cross-dataset than within-dataset accuracy on PhysioNet, producing negative gaps—a consequence of the near-chance PhysioNet baseline rather than genuine transfer.

Table 2. Cross-dataset balanced accuracy and normalised generalisation gap (Eq. 1). P → B = PhysioNet → BCI2a; B → P = BCI2a → PhysioNet.

		P → B		**B → P**	
Feature	**Model**	Acc.	Gap	Acc.	Gap
Time Dom.	LR	0.563	−0.054	0.531	0.168
	SVM	0.500	0.044	0.513	0.133
	RF	0.505	0.009	0.514	0.125
Band Pow.	LR	0.581	−0.038	0.537	0.164
	SVM	0.526	0.006	0.512	0.142
	RF	0.522	−0.019	0.507	0.167
FFT	LR	0.572	−0.029	0.528	0.140
	SVM	0.536	0.006	0.510	0.142
	RF	0.514	0.012	0.499	0.120

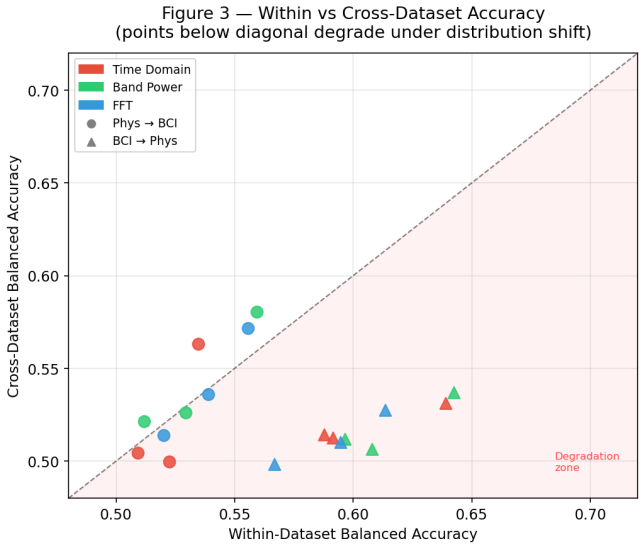

Figure 2. Within- vs. cross-dataset balanced accuracy. Points below the diagonal degrade under distribution shift. Symbols indicate direction; colours indicate feature set.

BCI2a → PhysioNet: gaps were substantially larger (0.12–0.17), representing *moderate* degradation by the interpretation guide. This reflects genuine performance loss when a model trained on 9 subjects is tested on 109 from a different recording system.

Averaged across all models and both directions, FFT showed the smallest mean normalised gap (0.065), followed by Band Power (0.070) and Time Domain (0.071). Logistic Regression was the most robust model (mean gap: 0.059). All values fall in the *small* degradation range, though the BCI → Phys direction alone is consistently moderate.

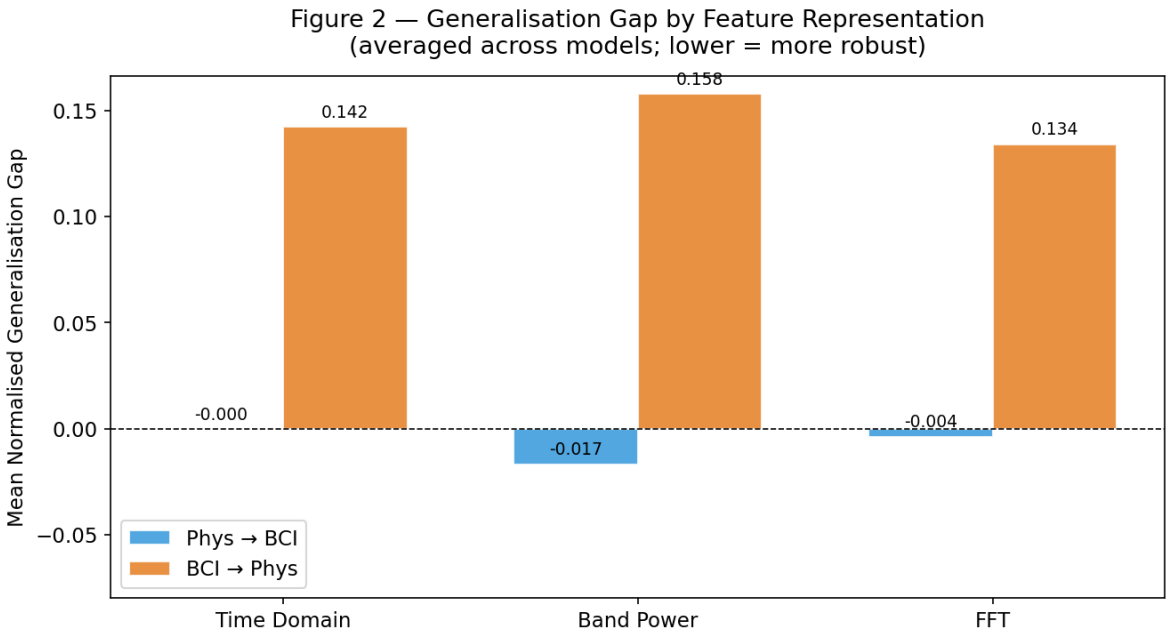

Figure 3. Mean normalised generalisation gap by feature representation, split by transfer direction (averaged across models). Lower values indicate more robust transfer. The BCI → Phys direction reveals the real degradation pattern; Phys → BCI gaps are obscured by the near-chance PhysioNet baseline.

V. Discussion

A. The Amplitude Distribution Shift

A diagnostic analysis of preprocessed arrays revealed a $9\times$ amplitude difference between datasets: PhysioNet signals ranged $\pm 856\,\mu$V while BCI2a ranged $\pm 96\,\mu$V. This arises from different recording hardware, electrode impedance, and amplifier gain—not from any difference in the underlying neural activity.

This is the primary mechanistic explanation for cross-dataset degradation. Time-domain variance features are proportional to amplitude squared, making them maximally sensitive. FFT magnitude features scale linearly with amplitude. Log-transformed band power compresses the scale difference: empirically, despite the $9\times$ amplitude gap, log band power means were nearly identical across datasets (PhysioNet: -22.8, BCI2a: -23.0).

B. Why FFT Outperformed Band Power

The hypothesis predicted band power as most robust, but FFT showed a slightly smaller average gap. Log-transformed band power demonstrably compresses the amplitude shift, yet FFT's finer 36-bin resolution retains discriminative spectral information that broad band averages lose. The StandardScaler inside the FFT pipeline normalises scale at training time, partially compensating for amplitude differences without the coarsening effect of band aggregation.

C. The Near-Chance PhysioNet Baseline

Within-dataset accuracy on PhysioNet was near chance (51–56%) across all configurations. This is not a pipeline error: run verification confirmed correct trial selection; signal diagnostics confirmed valid EEG-range amplitudes. The near-chance result reflects the genuine difficulty of motor imagery classification on an unscreened, heterogeneous 109-subject pool using zero-calibration classical features.

This has a direct implication for the literature. Reported accuracies of 80–90% [13, 8] on similar paradigms may reflect differences in evaluation protocol, subject selection, or subject-specific calibration rather than population-level cross-dataset generalisation. In this study, the within-dataset scores should be interpreted as pooled trial-wise cross-validation baselines rather than fully subject-independent population-level estimates, because trials from the same subject can appear in different folds.

D. The Asymmetric Transfer Directions

The two transfer directions told fundamentally different stories. Phys → BCI showed near-zero or negative gaps—not because transfer worked, but because the PhysioNet baseline was already at chance. BCI → Phys revealed moderate degradation of 12–17%, as 9 training subjects cannot capture the distributional variance of 109 test subjects from a different recording system. A study reporting only one transfer direction risks deeply misleading conclusions.

E. Logistic Regression as Most Robust Model

Logistic Regression showed the smallest average generalisation gap (0.059). This is consistent with the principle that simpler models generalise better under distribution shift: Random

Forest can overfit to dataset-specific patterns during training, amplifying degradation when those patterns are absent in the target.

VI. Limitations

Two datasets only. Findings are based on two datasets with similar left/right hand imagery paradigms. Generalisation to other EEG tasks (P300, SSVEP, cognitive load) or more diverse hardware cannot be assumed.

Near-chance PhysioNet baseline. Near-chance within-dataset accuracy on PhysioNet severely limits interpretability of the Phys $\rightarrow$ BCI direction: a meaningful generalisation gap requires a baseline sufficiently above chance.

Intentionally simple time-domain features. The time-domain family uses only per-channel mean and variance. This is an intentionally weak baseline that tests simple statistical time-domain features, not time-domain representations in general. Richer time-domain families (e.g., Hjorth parameters, zero-crossing rates, AR coefficients) exist and may show different robustness characteristics.

Classical models only. Deep learning may show qualitatively different generalisation patterns. This study deliberately excludes deep architectures to isolate feature representation effects; findings should not be extrapolated.

No domain adaptation. Results represent strict zero-shot transfer. This study establishes the unadapted baseline against which alignment and adaptation methods should be compared.

Single random seed. Cross-dataset experiments used a single seed. Within-dataset results include fold-level variance; cross-dataset point estimates lack confidence intervals.

No subject-level analysis. Population-level near-chance accuracy may mask a bimodal distribution of strong and weak motor imagery responders. Subject-level analysis remains for future work.

VII. Conclusion

This study provides a controlled empirical comparison of three classical EEG feature representations—time-domain statistics, log-transformed band power, and FFT magnitude—under strict zero-shot cross-dataset transfer between two publicly available motor imagery datasets. FFT features showed the smallest mean normalised generalisation gap (0.065), followed by band power (0.070) and time-domain features (0.071), with Logistic Regression as the most robust classifier. A $9\times$ amplitude difference between recording systems was identified as the primary mechanistic source of distribution shift. Log-transformed band power partially mitigates this shift, but differences between feature sets were modest, suggesting no classical representation fully solves the cross-dataset transfer problem. Within-dataset accuracy on an unscreened 109-subject population was near chance (51–56%) and, in this study, should be interpreted as a pooled trial-wise baseline rather than a fully subject-independent estimate. These findings highlight the importance of bidirectional cross-dataset evaluation as a standard practice and motivate future work on amplitude-invariant feature representations for robust neural decoding.

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